

Algorithms and Data Structures

26FS

Programme	BSc in Wirtschaftsinformatik BSc in Business Information Technology																
Degree	Bachelor																
ECTS	3																
Module coordinator	Prof. Dr. Patrik Christen																
Compulsory attendance	None																
Leading principle / Short description	Algorithms and data structures are at the heart of every computer program. Understanding and knowing common algorithms and data structures is therefore valuable for programmers. In this course we look at some basic linear data structures such as arrays, lists, stacks, and queues as well as some more advanced non-linear data structures such as trees and graphs. In both cases, common algorithms operating on the data structures are covered. A research case study on data structures in complex systems modelling and lectures on artificial intelligence round off the course. In addition, there are tutorials for each lecture implementing the treated data structures and algorithms. The course is provided online with video lectures, chat discussions, and individual coaching.																
Module content	1. Algorithms in Computing, Designing Algorithms, and Asymptotic Notation 2. Arrays, Matrices, Stacks, Queues, and Linked Lists 3. Representing Rooted Trees and Binary Search Trees 4. The Seven Bridges of Königsberg, Representations of Graphs, Breadth-First Search, and Applications of BFS 5. Research Case Study: Cellular Automata, Cellular Automata Data Structures and Algorithms, Programming Cellular Automata 6. Artificial Intelligence: Artificial Intelligence, Machine Learning, and Evolutionary Computation																
Competencies to be achieved	Knowledge and Understanding: Students ... ◦ know and understand common data structures and the related basic algorithms operating on these data structures. Applying Knowledge and Understanding: Students ... ◦ know to apply conceptual knowledge about data structures when implementing programs as well as conceptual knowledge about the related basic algorithms when implementing them. Making Judgements: Students ... ◦ have the ability to choose a suitable data structure and related algorithms for a given situation. Communications Skills: Students ... ◦ are able to explain common data structures and their related basic algorithms, also based on pseudocode.																
Prerequisites	Programming 1																
Teaching and learning methods	Online Lectures: Lecture, Presentation, Case Study Guided Self-Study: Individual work, Working with a partner																
Literature	T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein. Introduction to Algorithms, 4 ed., Cambridge, Massachusetts: The MIT Press, 2022.																
Remarks	None																
Grading	pass / fail																
Assessment	Assignment 50% <table> <tr> <td>Oral / Written</td><td>written</td></tr> <tr> <td>Timeframe</td><td>During the semester</td></tr> <tr> <td>Grading Scale</td><td>pass / fail</td></tr> <tr> <td>Remarks</td><td>Learning Journal</td></tr> </table> Assignment 50% <table> <tr> <td>Oral / Written</td><td>written</td></tr> <tr> <td>Timeframe</td><td>During the semester</td></tr> <tr> <td>Grading Scale</td><td>pass / fail</td></tr> <tr> <td>Remarks</td><td>Two-Page Essay</td></tr> </table>	Oral / Written	written	Timeframe	During the semester	Grading Scale	pass / fail	Remarks	Learning Journal	Oral / Written	written	Timeframe	During the semester	Grading Scale	pass / fail	Remarks	Two-Page Essay
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Module details**Algorithms and Data Structures (KW 8 - 15) - Sat - Windisch**

Time	8:15 AM - 12:00 PM
Language	Englisch
Max. participants	50
Periodicity	Weekly
Lecturers	Prof. Dr. Patrik Christen
Number	1-26FS.W-B-WIBIT-AlgDS_en.EN/WM

